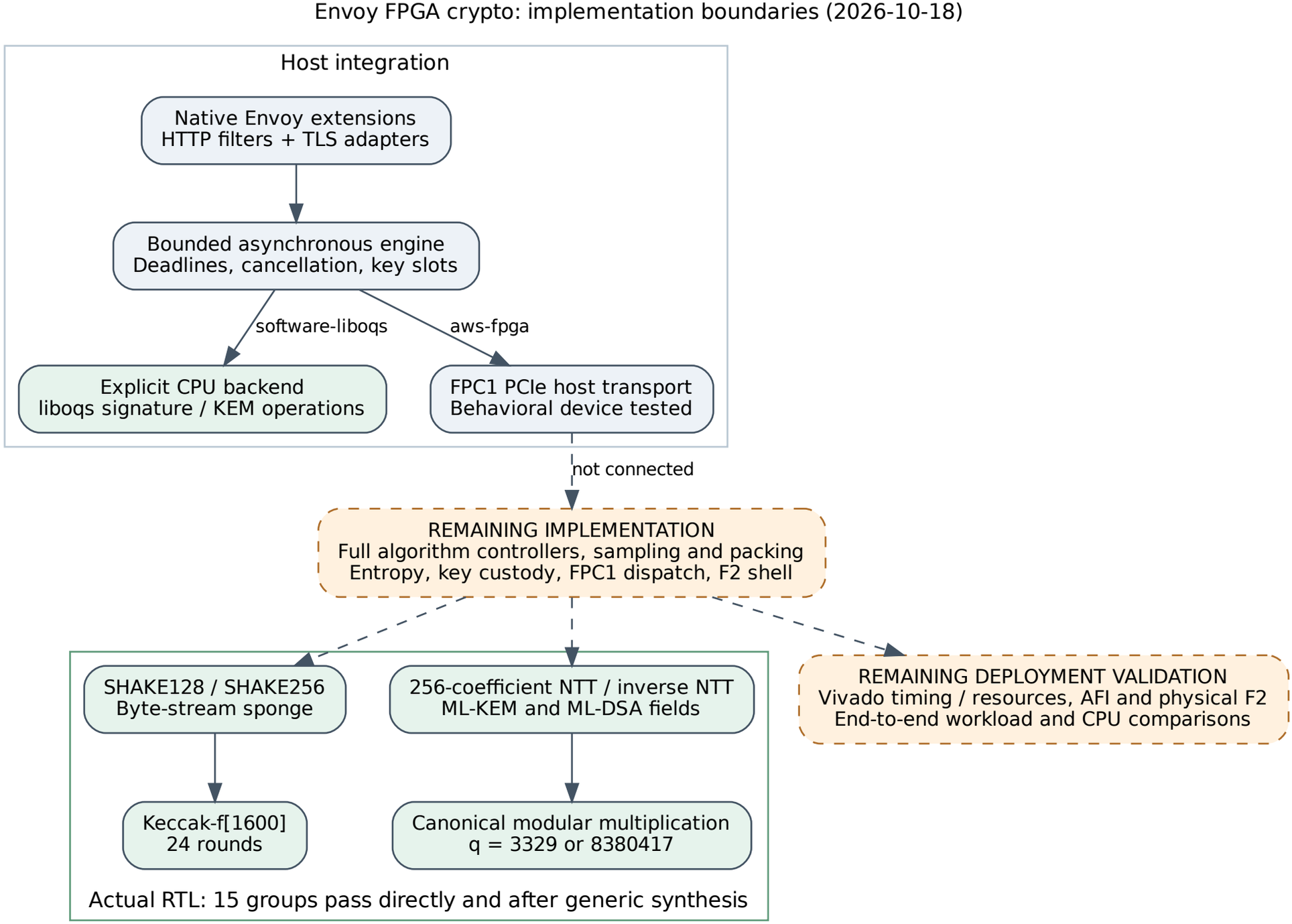
Envoy FPGA crypto: implementation boundaries (2026-10-18)
Host integration
Native Envoy extensions
HTTP filters + TLS adapters
Bounded asynchronous engine
Deadlines, cancellation, key slots
software-liboqs
aws-fpga
Explicit CPU backend
liboqs signature / KEM operations
FPC1 PCIe host transport
Behavioral device tested
not connected
REMAINING IMPLEMENTATION
Full algorithm controllers, sampling and packing
Entropy, key custody, FPC1 dispatch, F2 shell
SHAKE128 / SHAKE256
Byte-stream sponge
256-coefficient NTT / inverse NTT
ML-KEM and ML-DSA fields
REMAINING DEPLOYMENT VALIDATION
Vivado timing / resources, AFI and physical F2
End-to-end workload and CPU comparisons
Keccak-f[1600]
24 rounds
Canonical modular multiplication
q = 3329 or 8380417
Actual RTL: 15 groups pass directly and after generic synthesis